Digital Research Toolkit for Linguists

Week 14: AI for the humanities, version control, SSH

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July 8, 2025

Psycholinguistics and Cognitive Modeling Lab

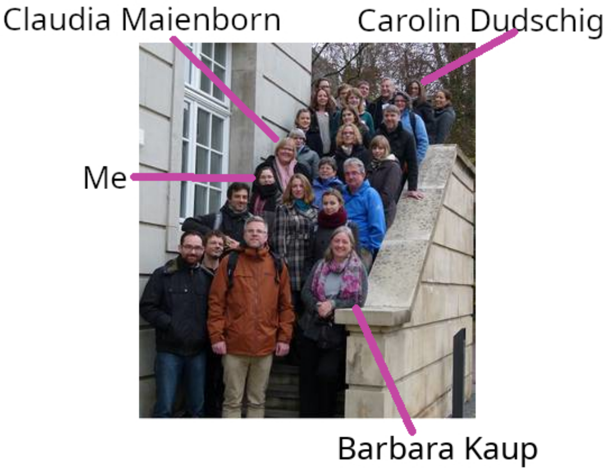
Homework

- Any issues installing git?

Homework

Goal: Learn to write concise scientific texts. Failed by 50%.

✖ **Dudschig, C.**, Mackowiak, K., **Maienborn, C.**, & **Kaup, B.** (2016). “The Moses illusion in different languages.” *Language and Cognition*, 8(2), 225–248.

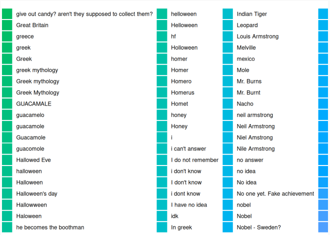


Reading comprehension

> Submit your report as a ZIP file containing:
>
> - a Quarto markdown file (QMD)
> - an HTML file
> - any additional files needed to generate the report (e.g. data, images)
>
> At least two plots.
>
> Add at least 10 relevant citations and print them in the bibliography.

Almost everyone missed at least one of these.

Quarto and Plots



`@tbl-foo` and `@fig-bar` → Table X and Figure X

```
1. Numbered list
    - Unnumbered list
```

Always double-check citations: `[@van1990]`

Speckmann, F. 2022. "Moses Illusion"
Elizabeth JM, et al. 2014. "Knowledge Neglect: Failures to Notice Contradictions with Stored Knowledge." January.
...and more!

Human or human-like?

Moses Illusions or semantic illusions occur when readers fail to recognize inconsistency in a text even if they were warned and know the correct word (Erickson and Mattson 1981).

Human

The Moses illusion is a well-documented semantic illusion in which people fail to notice a semantic error in a question and respond as if the question were correct.

Machine

Discourse Representation Theory (DRT) is a formal framework developed by Hans Kamp (1981) to model how linguistic meaning is constructed incrementally across connected sentences in discourse. At its heart, DRT proposes that understanding discourse involves building Discourse Representation Structures (DRSs)—mental models that represent entities, events, and relationships introduced in the discourse.

Machine

In formal linguistics, discourse representation theory (DRT) is a framework for exploring meaning under a formal semantics approach. One of the main differences between DRT-style approaches and traditional Montagovian approaches is that DRT includes a level of abstract mental representations (discourse representation structures, DRS) within its formalism, which gives it an intrinsic ability to handle meaning across sentence boundaries.

Human

Crash course in plagiarism

“The Moses illusion (also known as semantic illusion) was first identified by T.D. Erickson and M.E. Mattson in their article ”From Words to Meaning: A Semantic Illusion” (Journal of Verbal Learning and Verbal Behavior, 1981).” (Nordquist 2025)

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It was first identified by T.D. Erickson and M.E. Mattson in their article ”From Words to Meaning: A Semantic Illusion”, written in 1981.

The Moses illusion was first identified by T.D. Erickson and M.E. Mattson in their article ”From Words to Meaning: A Semantic Illusion” (Journal of Verbal Learning and Verbal Behavior, 1981).

Same for Wikipedia and my slides.

You have until Friday, July 11th at noon to re-submit last week's assignment. You will fail this assignment if you neither follow the instructions, nor give your best honest effort.

Questions?

Data → **Understand** → **Communicate** → **Share**

Data: create, find, data types, formats, encoding

Understand: R, packages, import, clean, transform, debug, visualize, export

Communicate: document, create clean and beautiful reports

Share: connect, collaborate, backup, replicate

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1. AI and LLMs
2. Version control
3. Git and GitHub
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AI and LLMs

Key Large Language Models

LLM	Creator	Strengths
ChatGPT	OpenAI	Versatile, general-purpose
Claude	Anthropic	Safe-ish, helpful, high context window
Gemini	Google DeepMind	Multimodal, Google integration
LLaMA	Meta AI	R&D, enterprise
DeepSeek AI	DeepSeek	Programming & scientific workflows Chinese company
Perplexity AI	Perpexity	Transparent, source-citation. GPT-4 + Claude + **?**
Grok	xAI	Twitter integration. Likely LLaMA + **?**

Notable AI tools

Academic

Elicit: AI for literature reviews and academic research

Julius: AI business assistant for data and analytics

Scite.ai: AI for studies that support or contradict claims

Programming

GitHub Copilot: AI pair programmer by GitHub + GPT

Tabnine: swift, privacy-respecting programming AI

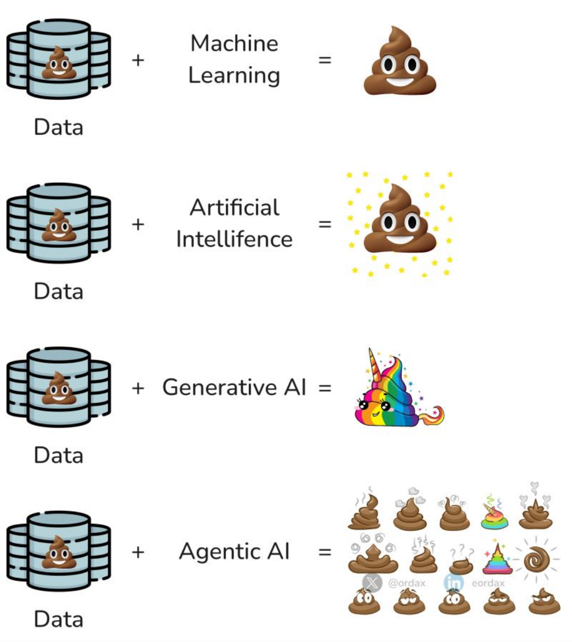
Cursor: VSC + GPT + Claude

Others

Image: DALL·E, Stable Diffusion, Midjourney...

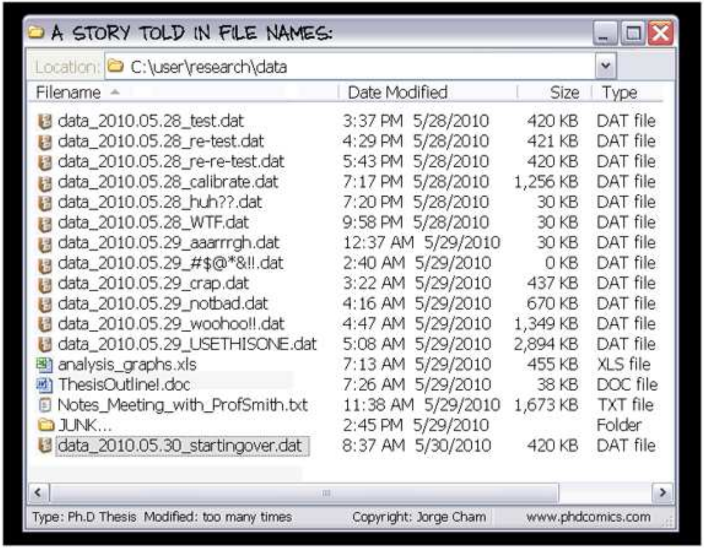
Video: Sora, Veo 3, Dream Machine, Gen-2...

Quality data and quality user

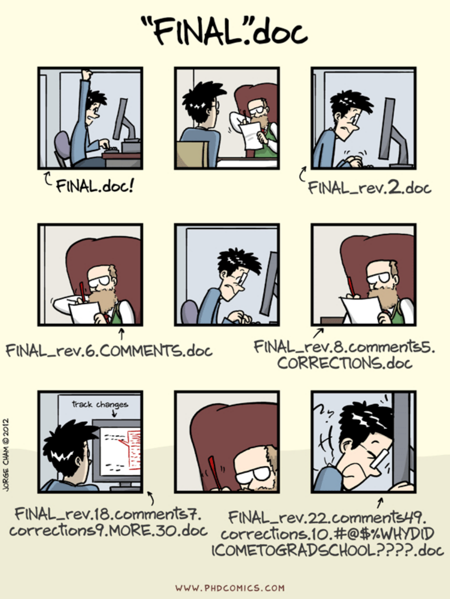


Version control

Does this look familiar?



Does this look familiar?



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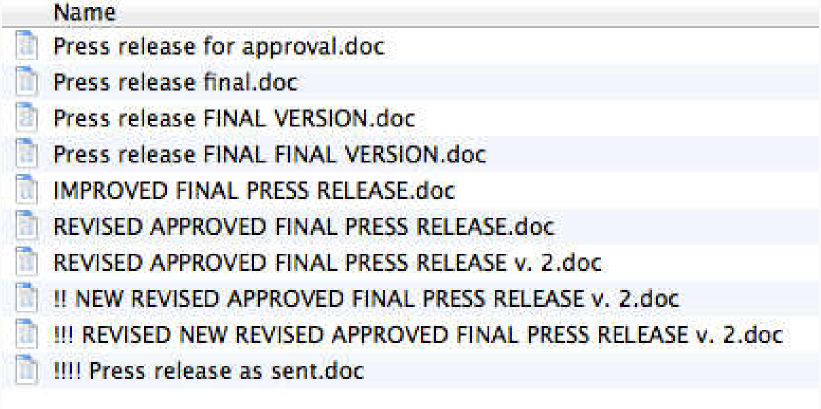
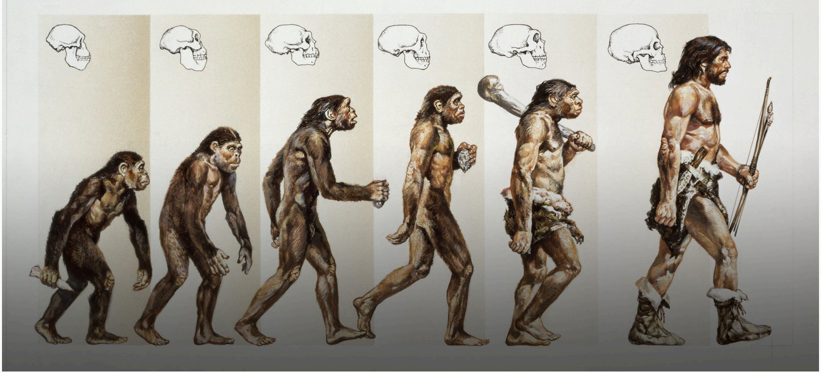


Popular choices: date, initials, version nr, final, copy, reviewed

Renaming, saving as, adding initials etc. to file names works for small personal project but breaks down with collaboration, big projects.

Version control

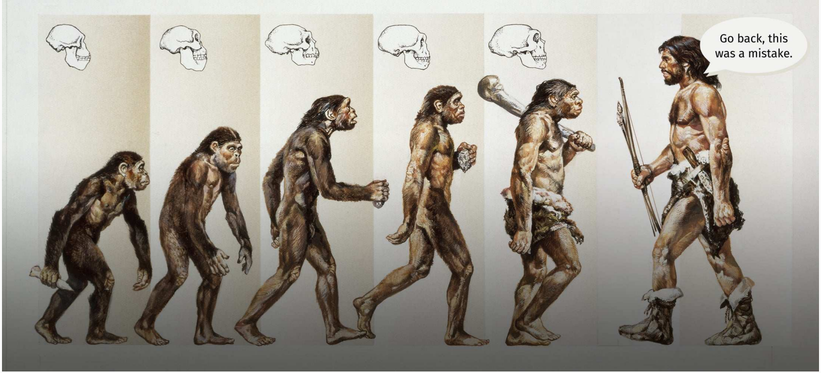


Gettyimages

Version control helps you track changes over time.

Version control

Z



Gettyimages

When you mess up you can **go back** to when things worked.

Why use version control?

History and tracking

Keeps a detailed record of all stages of the project. Allows tracking who made which change and why.

Documentation

Documents changes, reasons for changes, and related discussions.

Backup and restore

A backup for the project. In case of mistakes, data loss, or sabotage, the previous version(s) can be easil restored.

Why use version control?

Experimentation

You can try out new ideas without affecting the core project. Branches can be safely created, removed, or integrated as needed.

Collaboration

Multiple people can work on the same project simultaneously and merge their changes without many conflicts. Different groups can work on different parts of a project.

Remote work

Many teammates can work from different locations and on different machines with different OS and synchronize their work.


How tf did they build this
without any
version
control

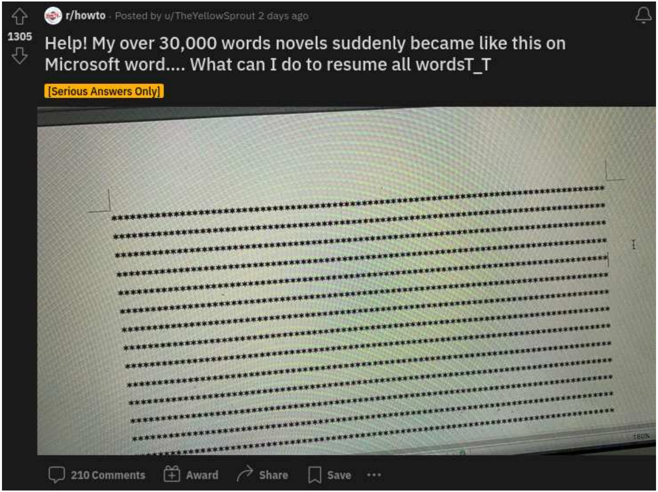
Tracking changes

`diff` is great locally and you can use DiffChecker
$\rightarrow$ paid + online + data security?

Multiple people working on the same project in parallel makes files messy, fast:

- Collaborator 1 is writing the intro
- Collaborator 2 is adding the statistics
- Collaborator 3 is summarizing the results
- Proofreader is checking for errors
- ...

Backup



Remember to backup important files locally and in the cloud!

Git and GitHub

Version control and git

Why learn Git?

- Keep track of your data
- Synchronize across devices
- Collaborate effectively
- Resolve errors and fix mistakes
- Parallelize processes
- Backup and store files
- Remember, revert, and restore (short- and long-term)
- Be more employable as (data) scientists and developers
- It's free and open source


THIS IS GIT. IT TRACKS COLLABORATIVE WORK ON PROJECTS THROUGH A BEAUTIFUL DISTRIBUTED GRAPH THEORY TREE MODEL.
COOL. HOW DO WE USE IT?
NO IDEA. JUST MEMORIZE THESE SHELL COMMANDS AND TYPE THEM TO SYNC UP. IF YOU GET ERRORS, SAVE YOUR WORK ELSEWHERE, DELETE THE PROJECT, AND DOWNLOAD A FRESH COPY.

Git

(slang: worthless, foolish, unpleasant person) an OS version control software that works locally on your computer. Great for individual use.

GitHub

the most popular code hosting platform that makes collaborating using git easier. LinkedIn and Twitter for developers, owned by Microsoft. Uni Stuttgart has their own GitHub server `https://github.tik.uni-stuttgart.de/`

GitLab

an alternative code hosting platform that has somewhat different user permissions and includes Continuous Integration (no need to pull → this will be explained soon).

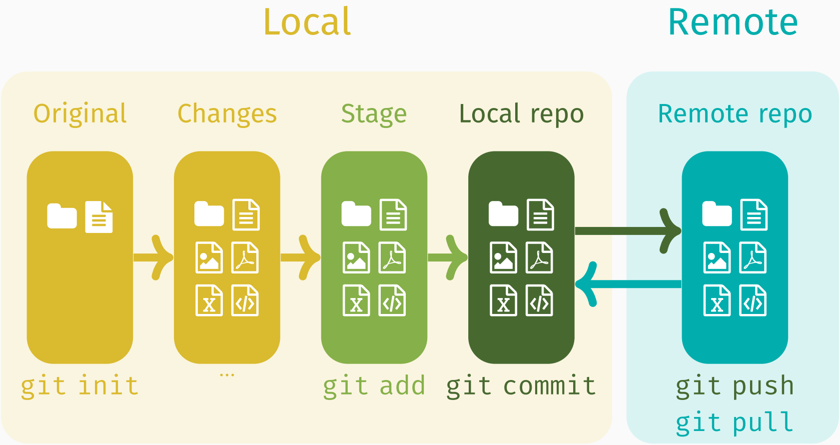
Git vs GitHub

git A tool for tracking changes to your files on **your computer**.

A service for hosting & sharing those files **online** + collaboration tools.

git Git	GitHub
Version control system	Web-based platform
Program	Service that uses Git
Keeps track of file changes over time	Hosts your Git repositories online
Local to you	Remote, makes collaboration possible
Takes snapshots (commits) of a project that you can look at	You push your local repositories + commits to a server
Managing your code and a local copy of someone else's code	Collaboration across multiple remote machines
Offline on your computer	On a server in the cloud
Not social media	LinkedIn for programmers

Workflow


Local
Remote
Original
Changes
Stage
Local repo
Remote repo
git init
...
git add
git commit
git push
git pull

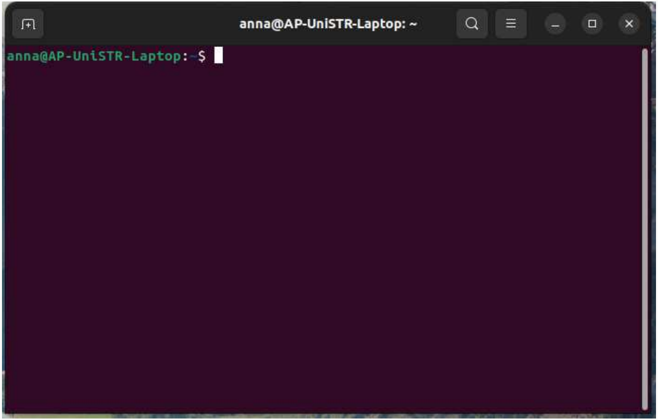
Command line and SSH

Terminal, console, shell, or command line?

Lat. *terminus* = end, boundary (bus, airport, battery, patient, ...)

A **terminal** is a text input and output environment (e.g. command prompt, gnome-terminal). It used to be physical device, located at the *end of the cable* to a computer.

Terminal, console, shell, or command line?

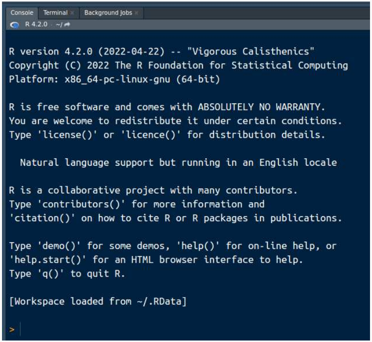


A **terminal window** (or terminal emulator) is a text-only window that emulates a terminal/console in a GUI. It runs a shell.

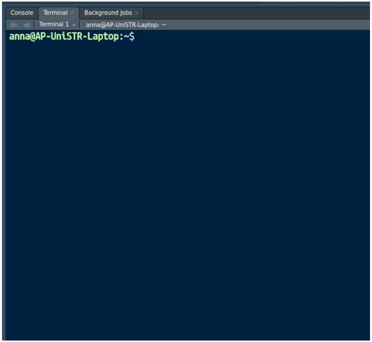
Terminal, console, shell, or command line?

A **console** is (generally) a type of physical terminal. It is sometimes synonymous with a terminal (e.g. for direct communication at a low level with the OS). Terminal vs. console = rap vs. hip hop

Console vs Terminal in RStudio

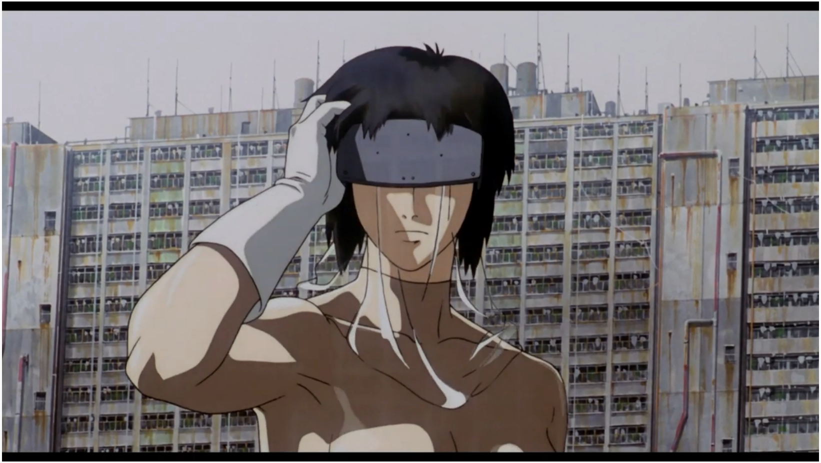


Console → R code



Terminal → access to system shell

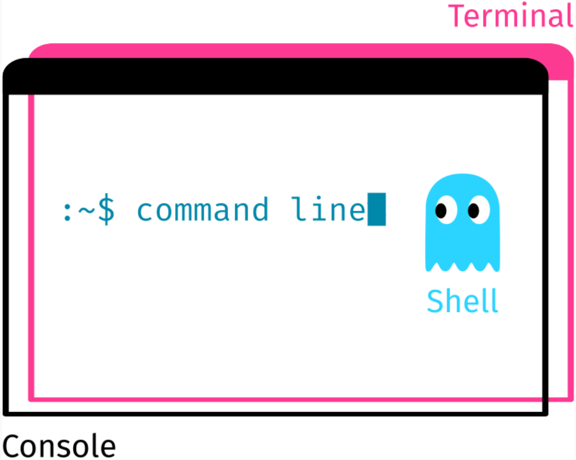
Terminal, console, shell, or command line?



A **shell** is a command-line interpreter (e.g. Bash, Power Shell, CMD). It's a program that runs in the terminal. It reads and processes commands, and outputs the results.

Terminal, console, shell, or command line?

A **command line** is an area to the right of the command prompt where you type commands.



The command line is what you input into your shell. The shell is what processes and executes the command. The terminal is the window in which the shell runs, which is often the same as the console.

Source

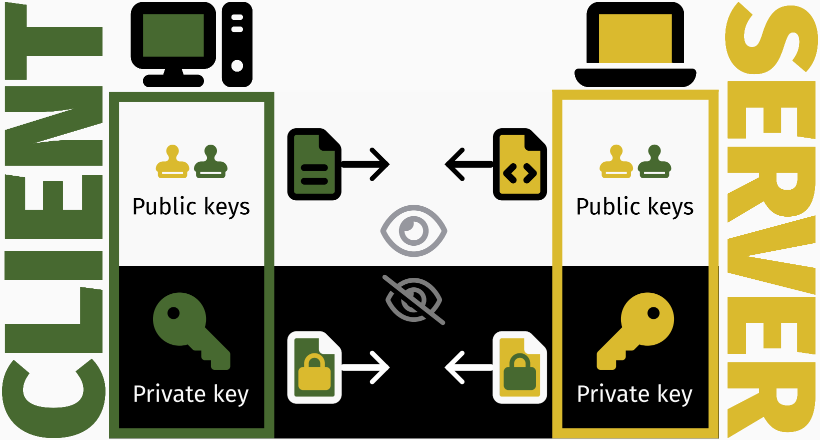
SS Hwæt?

SSH = Secure Shell Protocol

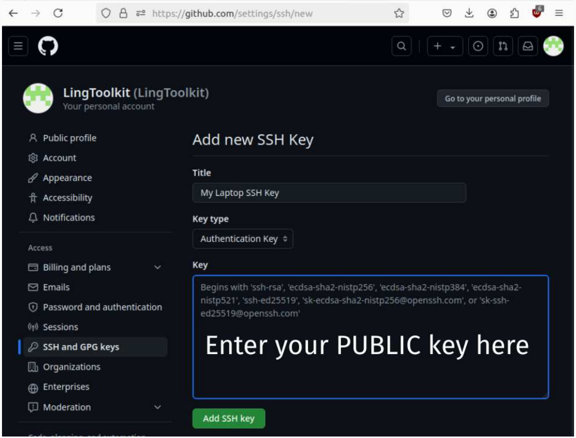
A 🐚 is a command-line interpreter that runs in the terminal.

SSH is a network protocol for **operating services securely over an unsecured network**, e.g. remote login and command-line execution (connect and authenticate). (Wikipedia 2024)

Client, server, and asymmetric encryption



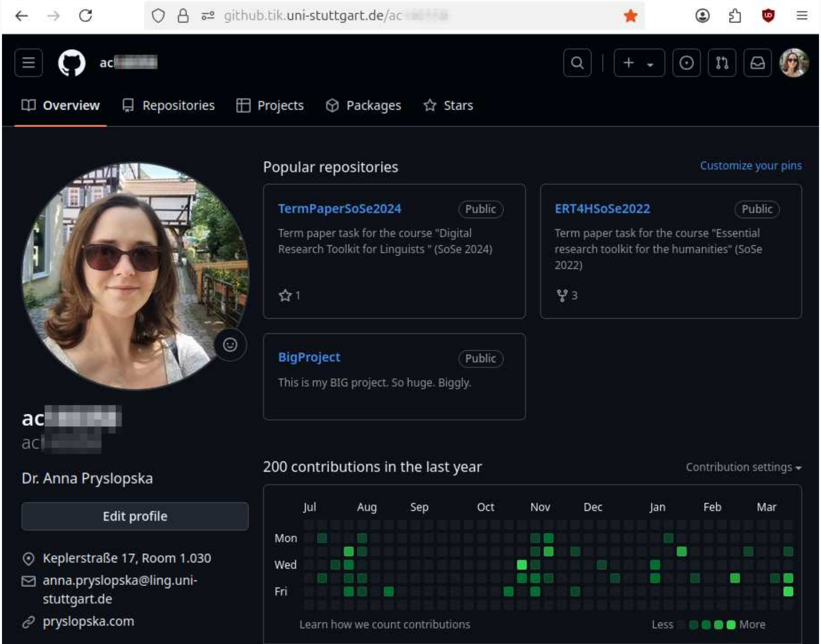
Practically, we're using SSH as a substitute for passwords in communicating with GitHub.



Exercise 1: Make a university GitHub account

`github.tik.uni-stuttgart.de`

GitHub



github.tik.uni-stuttgart.de/ac
ac
Overview
Repositories
Projects
Packages
Stars
Popular repositories
Customize your pins
TermPaperSoSe2024
Public
Term paper task for the course "Digital Research Toolkit for Linguists " (SoSe 2024)
1
ERT4HSoSe2022
Public
Term paper task for the course "Essential research toolkit for the humanities" (SoSe 2022)
3
BigProject
Public
This is my BIG project. So huge. Biggly.
ac
ac
Dr. Anna Pryslopska
Edit profile
Keplerstraße 17, Room 1.030
anna.pryslopska@ling.uni-stuttgart.de
pryslopska.com
200 contributions in the last year
Contribution settings
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Mon
Wed
Fri
Learn how we count contributions
Less
More

Exercise 2: Create a readme file

`readme.md`

Readme.md

- Documentation
- First Contact
- What, how, why?

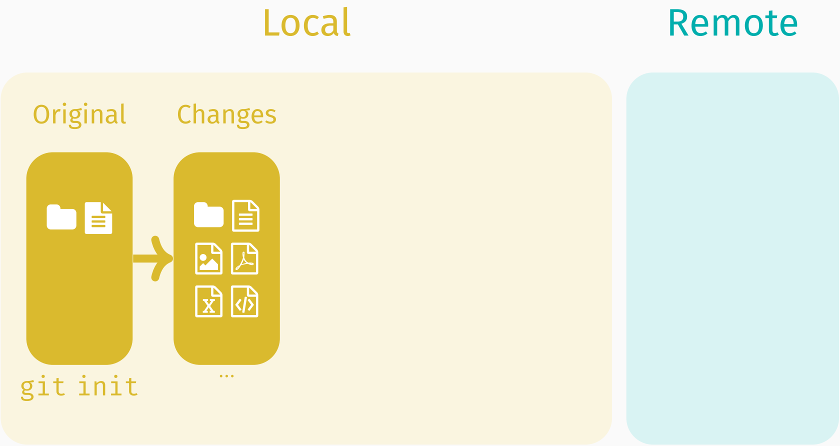
Open VSC, write a Readme file, save it as `readme.md` in your project folder.

```
# Readme template project
A nice & descriptive name for the project

## About The Project
A brief overview, e.g. the abstract.

## Installation / Requirements / Usage
Usually separate, but I've pooled them.

- What software is needed?
- What packages are needed?
- How to run the code?

## Structure
How are the files & folders organized?

## Contributors
Who am I? My name is Ned.
```

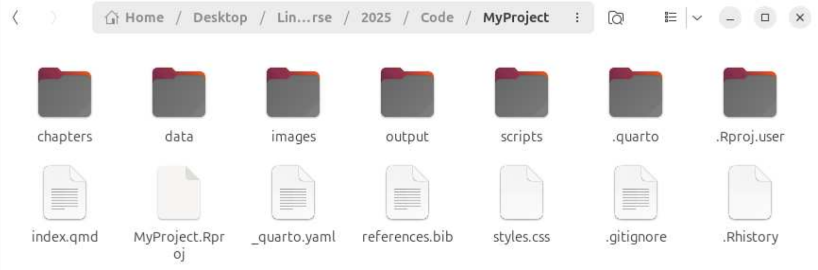
Exercise 3: Create a git repository

`git init`

Workflow



Initialize git



right click in folder (→ show more options) → “Git Bash Here” OR “Open in Terminal”

click in folder → Finder → Services → New Terminal at Folder

right click in folder → “Open in terminal”

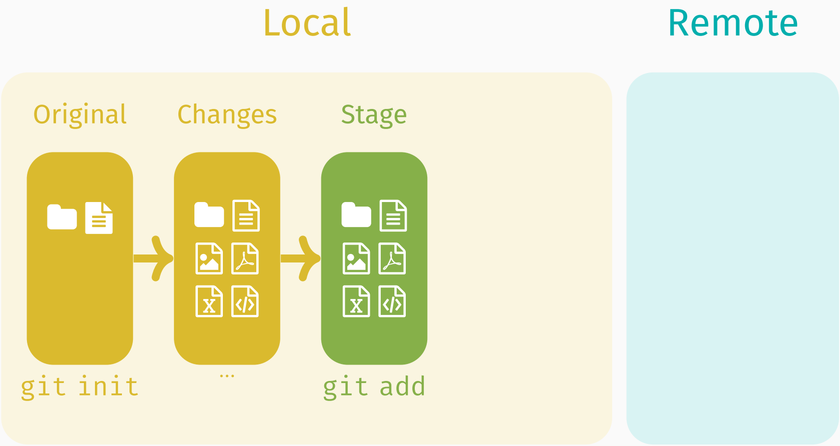
Type `git init` and press Enter

Exercise 4: Stage the readme file

```
git status
```

```
git add readme.md
```

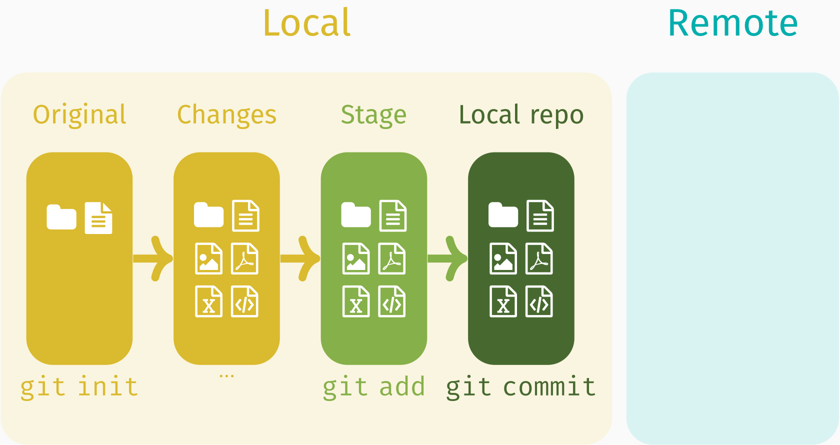
Workflow



Exercise 5: Commit the changes

```
git status
git commit -m "First commit"
git status
```

Workflow



Local
Remote
Original
Changes
Stage
Local repo
X
X
X
git init
...
git add
git commit

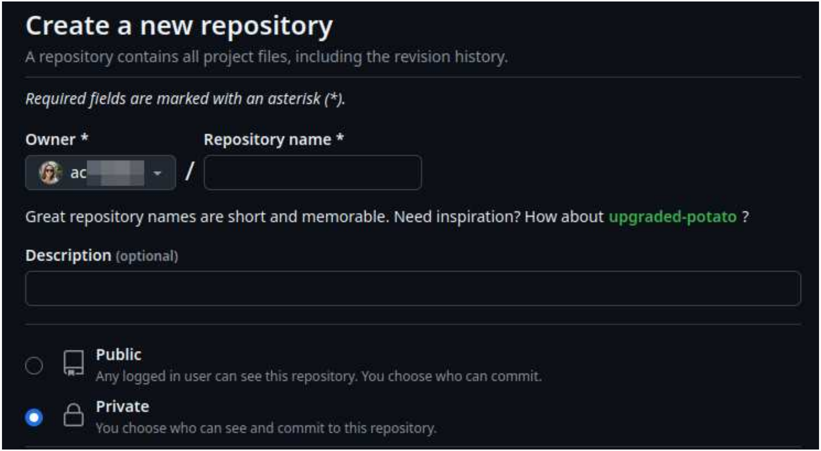
Exercise 6: Make a new SSH key pair and add it to GitHub.

```
ssh-keygen -t ed25519
```


DIGITAL
RESEARCH
SSH FOR GITHUB
ANNA PRYSLOPSKA
TOOLKIT FOR
LINGUISTS

Exercise 7: Create a new repository

`github.tik.uni-stuttgart.de/new`


Create a new repository
A repository contains all project files, including the revision history.
Required fields are marked with an asterisk (*).
Owner *
ac
/
Repository name *
Great repository names are short and memorable. Need inspiration? How about upgraded-potato ?
Description (optional)
Public
Any logged in user can see this repository. You choose who can commit.
Private
You choose who can see and commit to this repository.

Questions?

Summary

- ✔ Key LLMs and AI tools for research
- ✔ Version control
- ✔ Git basics, GitHub and SSH
- ⏩ More git and GitHub

Read more

Git guide: `https://github.com/git-guides/`

Another git guide:
`http://rogerdudler.github.io/git-guide/`

Git tutorial: `http://git-scm.com/docs/gittutorial`

Another git tutorial: `https://www.w3schools.com/git/`

Git cheat sheets: `https://training.github.com/`

Ask questions: `https://stackoverflow.com`

Homework assignment

Homework assignment due July 14th at 12:00

- Complete assignment 12 ($\rightarrow$ ILIAS) As part of the assignment, you will:
  - Finish adding your key to the keychain (see forum post about troubleshooting).
  - Add your public key to your GitHub profile.
  - Create a new repository on GitHub and add me as a collaborator.

Homework assignment correction

If needed, update last week's homework by noon on Friday, July 11th. Pay attention to the instructions for the report (e.g. number of figures, relevance of plots, hyperlinking, bibiography, what files to upload and how). As a reminder:

Assignment 10 (+ `index.qmd` file)

Create a report on the Moses illusion experiment from this seminar. I've provided the example file index.qmd as a guideline. Submit your report as a ZIP file containing: a Quarto markdown file, an HTML file, any additional files needed to generate the report (e.g. data, images)

Assignment 11

Add at least 10 relevant citations and print them in the bibliography (at least 1 book or book chapter, 1 article, 1 software or package). Print the bibliography from a .bib file.

Add another 5 sentences to the introduction (e.g. as a literature overview), another 5 sentences to the discussion, another 3-4 sentences to the remaining sections

Exam date

Tuesday, July 22nd 11:30–13:00, Kepler 17 (K2), Room M 17.81 (same time, weekday, and room)

Second exam date

Thursday, September 25th 11:30–13:00, Kepler 17 (K2), Room M 17.14 (same time, different weekday and room!)

Poster

You will receive new data on the Moses illusion. Task: clean, analyze, summarize, do literature research, make a scientific poster (+ give a short poster presentation)

References

Nordquist, Richard (May 2025). ***The Moses (Semantic) Illusion: Definition and Examples in Grammar.*** https://www.thoughtco.com/what-is-the-moses-illusion-1691328. ThoughtCo; updated May 8, 2025; accessed July 4, 2025.

Wikipedia (2024). ***Secure Shell.*** Accessed: 2024-07-12. URL: https://en.wikipedia.org/wiki/Secure_Shell.